(ii) Mechanism of enzyme action :

The enzyme-substrate union results in the release of energy. It is this energy which, in fact, raises the energy level of the substrate molecule, thus inducing the activated state (Figure 9.2). In this activated state, certain bonds of the substrate molecule become more susceptible to cleavage.

Evidences Proving the Existence of an ES Complex: The existence of an ES complex during enzymatically catalyzed reaction has been shown in many ways :

1. The ES complexes have been directly observed by electron microscopy and x-ray crystallography.
2. The physical properties of enzymes (solubility, heat sensitivity) change frequently upon formation of an ES complex.
3. The spectroscopic characteristics of many enzymes and substrates change upon formation of an ES complex. It is a case parallel to the one in which the absorption spectrum of deoxyhemoglobin changes markedly, when it binds oxygen or when it is oxidized to ferric state.
4. Stereospecificity of highest order is exhibited in the formation of ES complexes. For example, D-serine is not a substrate of tryptophan synthetase. As a matter of fact, the D-isomer does not even bind to the enzyme.
5. The ES complexes can be isolated in pure form. This may happen if in the reaction, $A + B \rightarrow C$, the enzyme has a high affinity for the substrate A and also if the other reactant B is absent from the mixture.
6. A most general evidence for the existence of ES complexes is the fact that at a constant concentration of enzyme, the reaction rate increases with increase in the substrate concentration until a maximal velocity is reached.

9.3. Enzyme Inhibition

A substance which binds with an enzyme and brings about a decrease in catalytic activity of that enzyme is called an enzyme inhibitor. This process is known as enzyme inhibition. The enzyme inhibitor may be organic or inorganic in nature. Enzyme inhibition is classified into three types as follows :

1) Reversible inhibition
2) Irrerversible inhibition.
3) Allosteric inhibition

9.3.1. Reversible inhibition :

When the inhibitor binds noncovalently with the enzyme and the inhibitor dissociates rapidly from the enzyme inhibitor complexes, it is known as reversible inhibition. It is further subdivided into

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